

	Work Instruction	Doc. Revision	02
	AXI V810 Y-Axis Read Head Encoder Removal & Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 July 2014	First Release	Muhammed Shameem
01	9 July 2015	Second Release	Muhammed Shameem
02	2 Oct 2016	Template Changes & Content update	ONG KEAN SHENG

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Parts Requirement

V810 Standard	V810 XXL	V810 S2EX
1. Allen Key (Metric Size)	1. Allen Key (Metric Size)	1. Allen Key (Metric Size)
2. Blue Spacer	2. Blue Spacer	2. Blue Spacer
3. 4XASP-A038	3. 4XASP-A096	3. 4JASC-A081
Y-Axis Linear Encoder	XXL Y-Axis Linear Encoder	S2EX Y-Axis Linear Encoder
Read Head Kit	Read Head Kit	Read Head Kit

Y-axis Read Head Encoder Removal

1. Turn off the X-Ray and SSCA Power.
2. Open left outer barrier

Remarks:

I. For Standard V810 Machine

- i. Unplug all the cables shown in the Figure 1

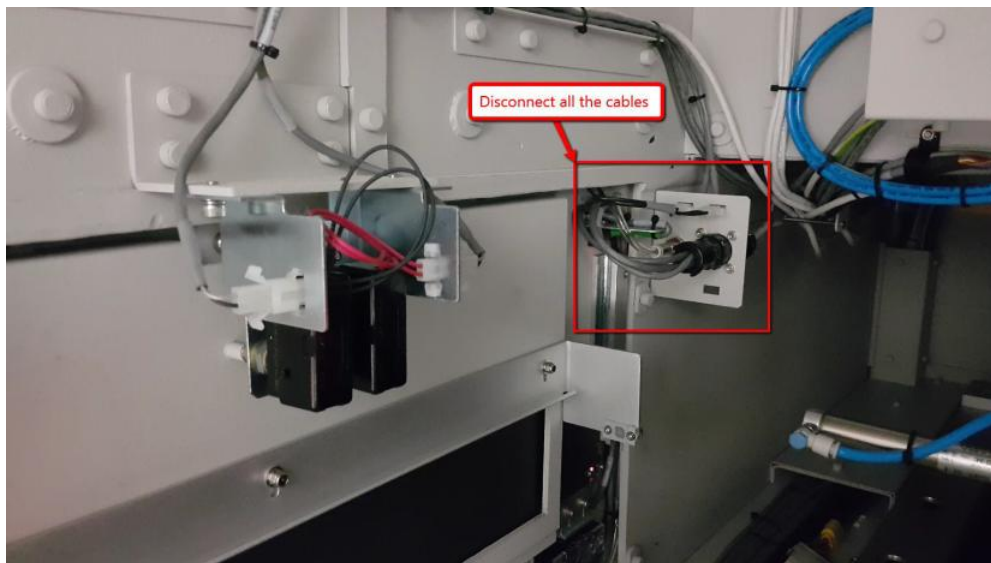


Figure 1 Left Outer Barrier Connectors

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- ii. Remove 4 screws as shown in the Figure 2 & remove outer barrier

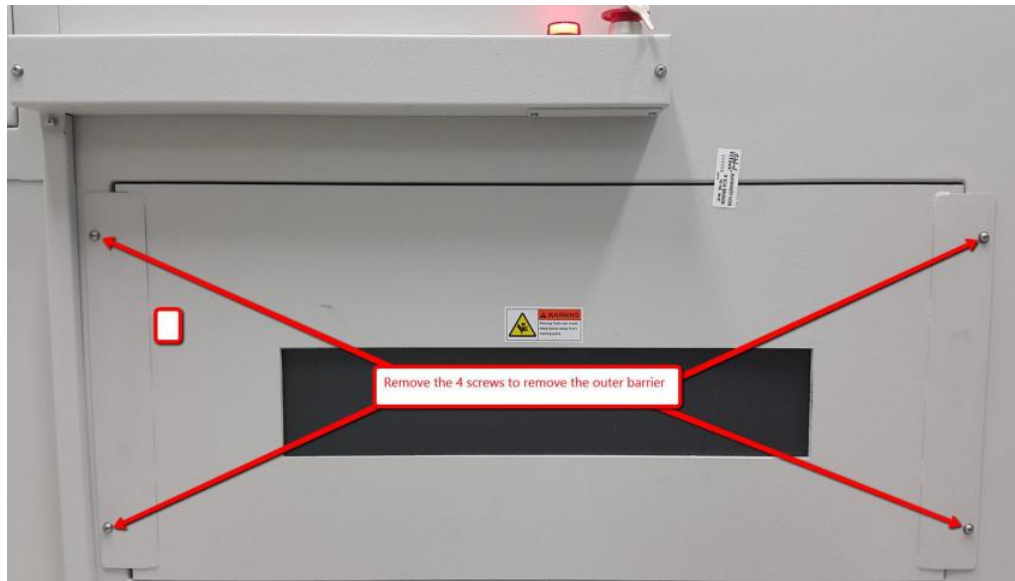


Figure 2 Left Outer Barrier

- iii. Remove left outer barrier

II. For S2EX and XXL machine

- i. Remove the 2 screws shown in the Figure 3



Figure 3 Left Outer Barrier

- ii. Open the left outer barrier to access the stage

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3. Open rear access door
4. Open 2x screws and remove the lead cover as shown in the Figure 4

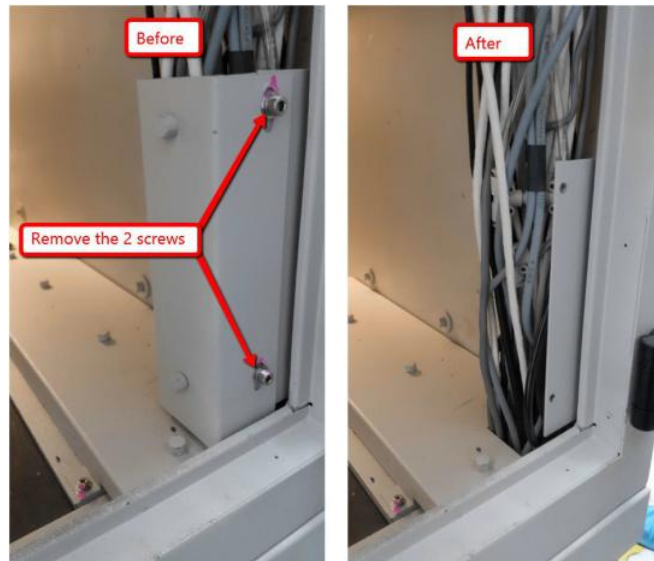


Figure 4 Lead Cover

5. Open 3 x screws and remove the lead cover as shown in the Figure 5

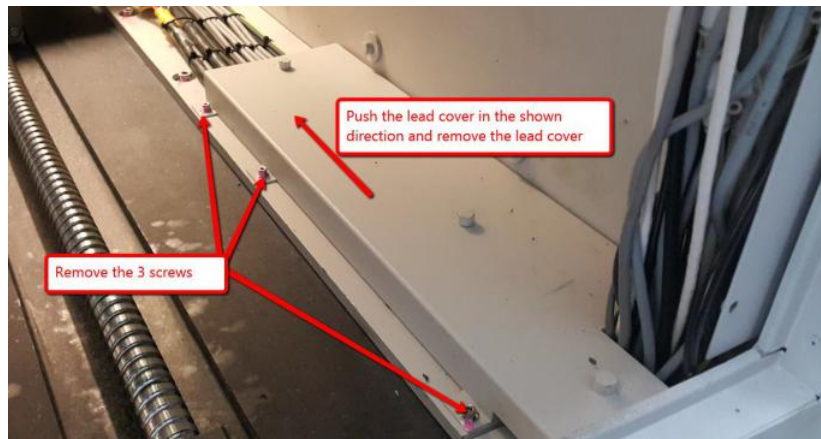


Figure 5 Lead Cover

6. Remove 2 x screws from the SSCA Cable Clamp as shown in the Figure 6

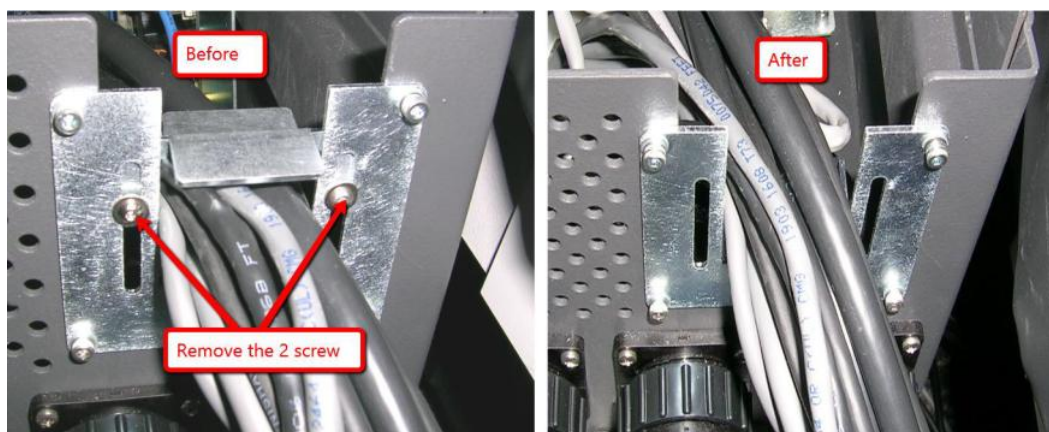


Figure 6 SSCA Cable Clamp

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7. Remove Y-Axis Read Head Encoder connector from Y-Axis Kollmorgen as shown in the Figure 6

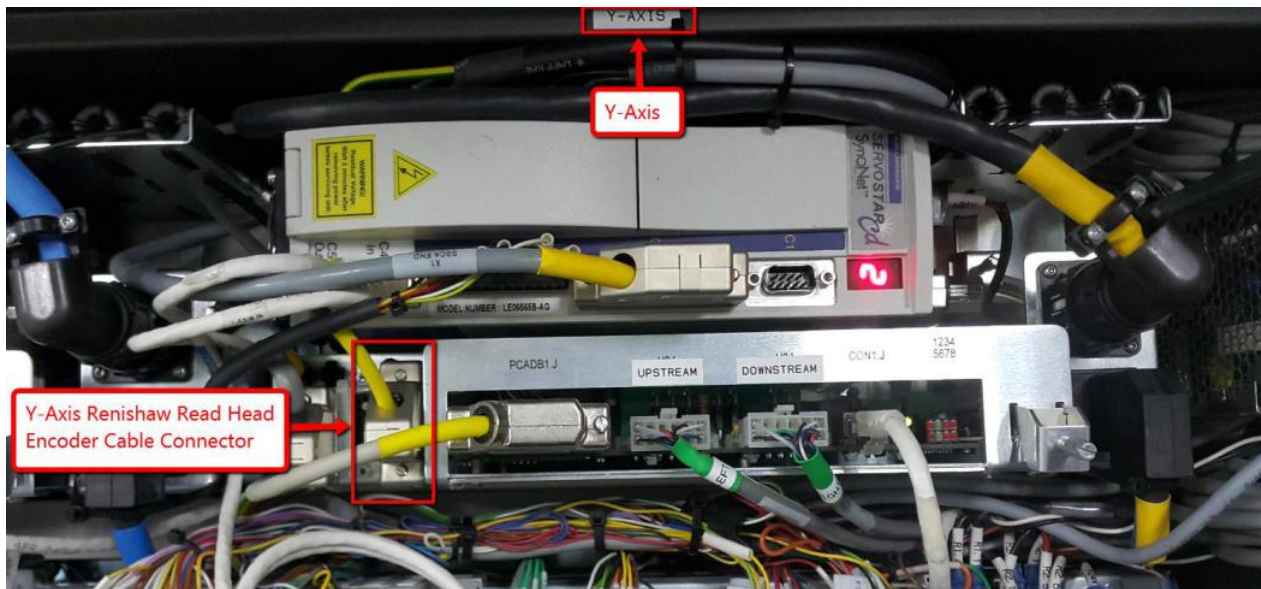


Figure 7 Y-Axis Read Head Encoder

8. Remove SHCS M5x20 (2x) to remove the Y-Axis Read Head Encoder Bracket as shown in the Figure 8



Figure 8 Read Head Encoder Bracket

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9. Remove the SHCS M3x20 (2X) to remove the Read Head Encoder from the Read Head Encoder Bracket as shown in the Figure 5

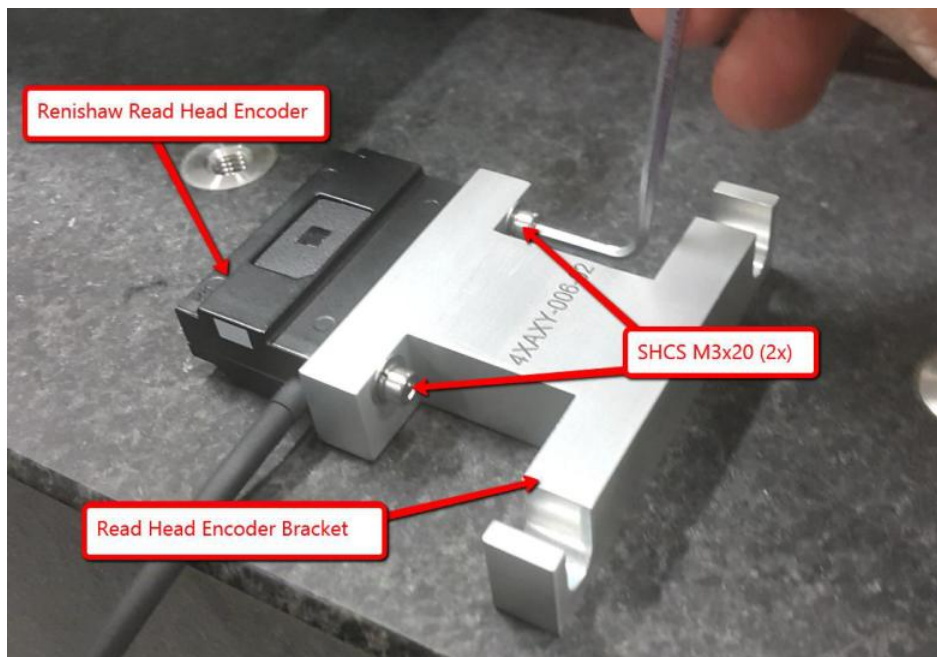


Figure 9 Y axis Read Head Encoder

10. Remove Y axis Read Head Encoder by refer routing which illustrated from Figure 10 to Figure 12
Remarks: Follow the Red Arrow for removal direction and necessary cable ties would require to remove

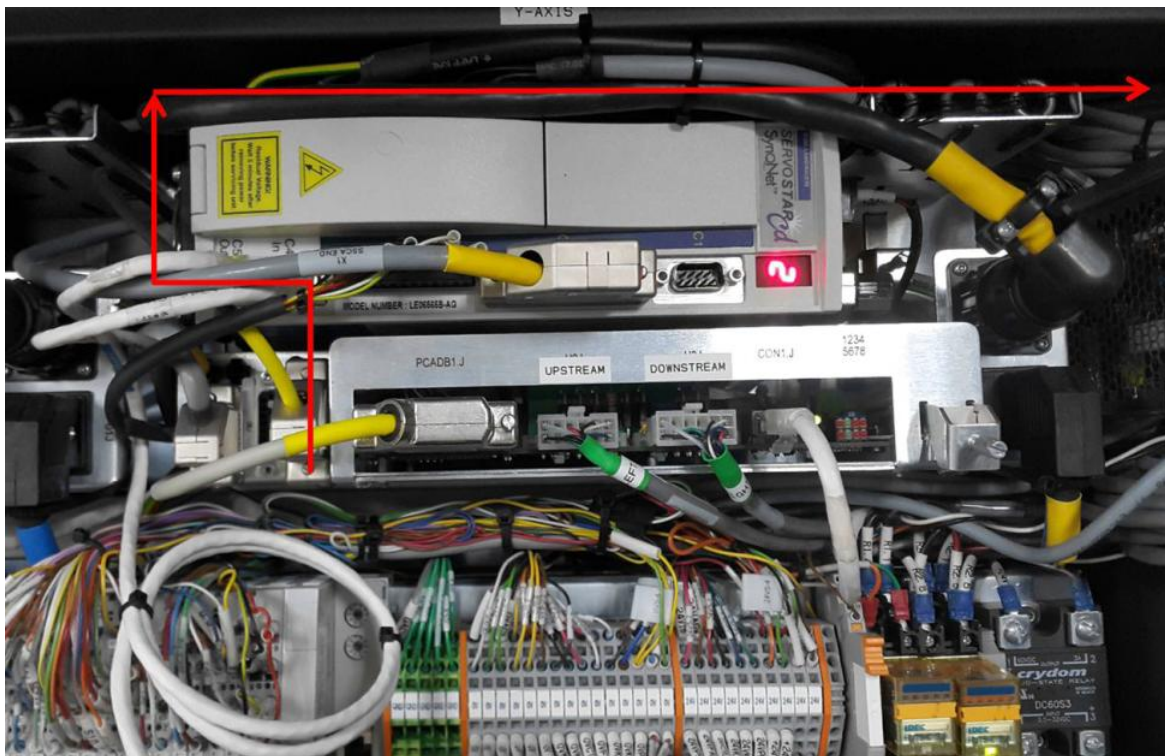


Figure 10 SSCA Y-Axis

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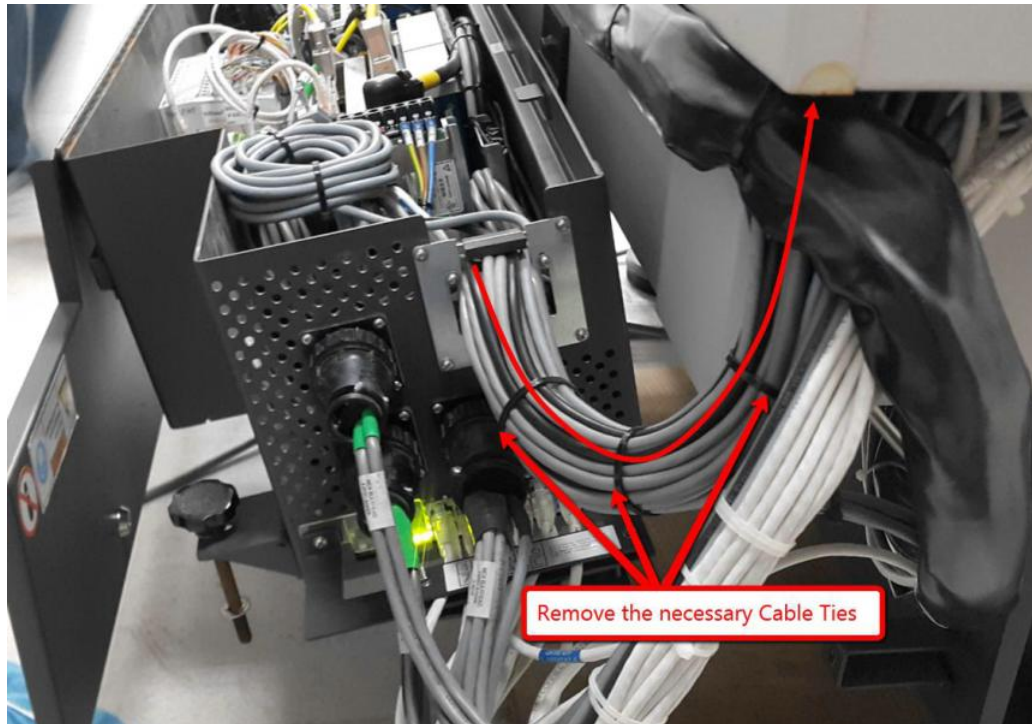


Figure 11 SCA Y-Axis Read Head Encoder

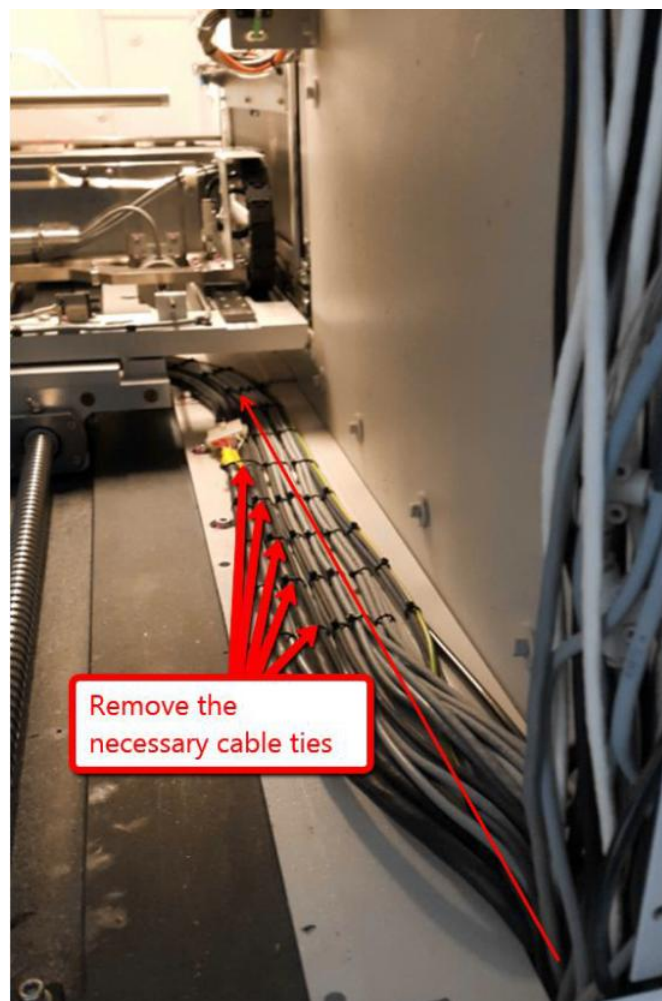


Figure 12 Cable Routing

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Y-axis Read Head Encoder Replacement

11. Prepare the new Y axis Read Head Encoder and ensure the items shown in Figure 13 is available for part replacement.

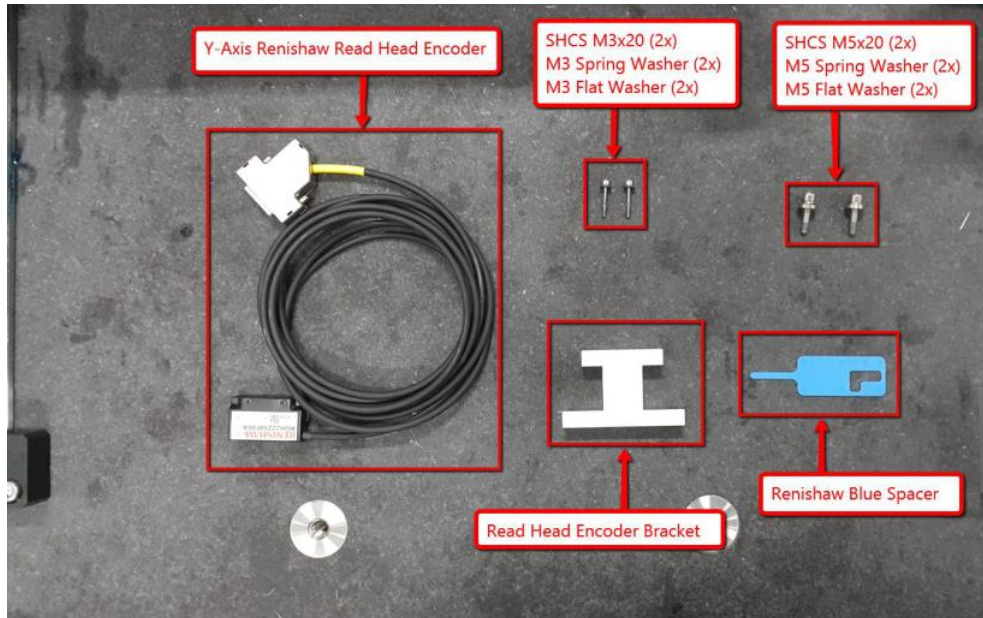


Figure 13 Replacement Parts

12. Assemble new Y-Axis Read Head Encoder to the Read Head Encoder Bracket as shown in Figure 14 by using SHCS M3x20 (2x)

Remarks :Y-Axis Read Head is require to push maximum in the shown direction as Figure 15 before tighten up.

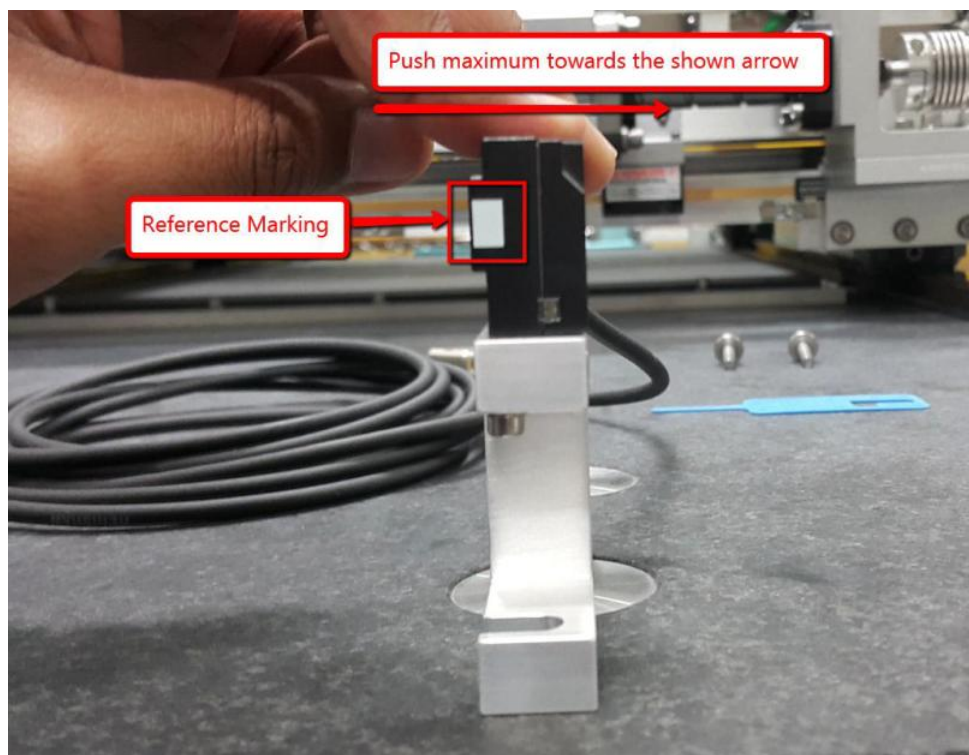


Figure 14 Read Head Encoder

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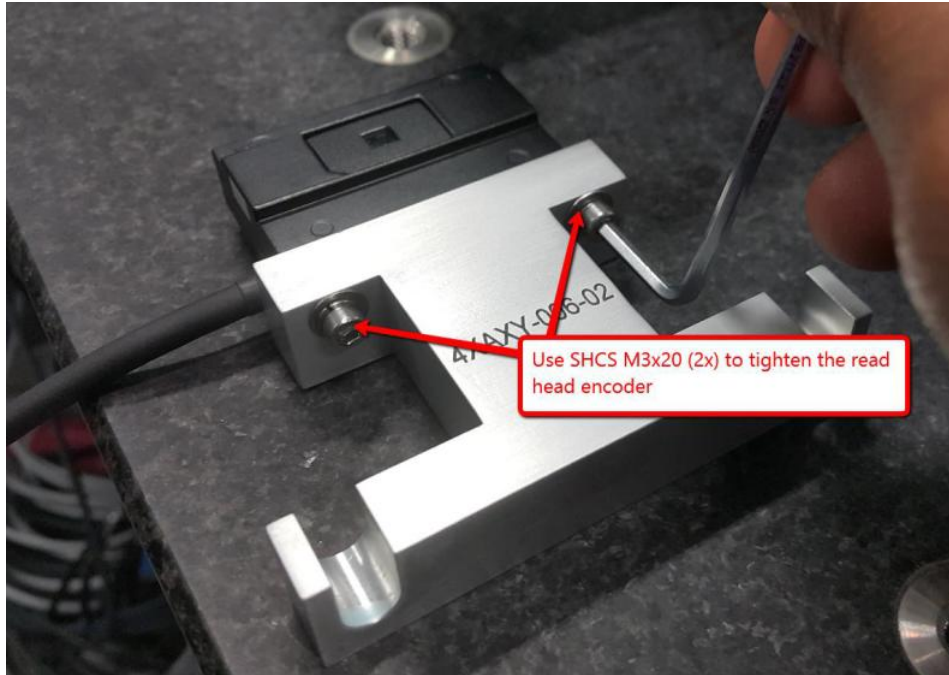


Figure 15 Read Head Encoder Assembly

13. Install the Read Head Encoder Bracket on the machine using the SHCS M5x20 (2x) as shown in the Figure 16



Figure 16 Read Head Encoder

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14. Route the cable as shown in Figure 17 to Figure 20

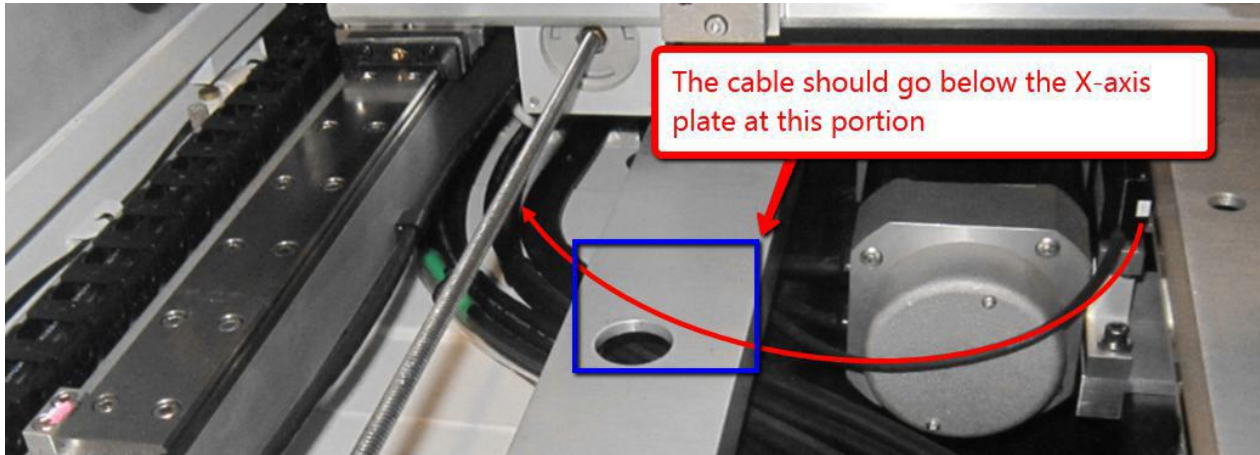


Figure 17 Read Head Encoder Routing

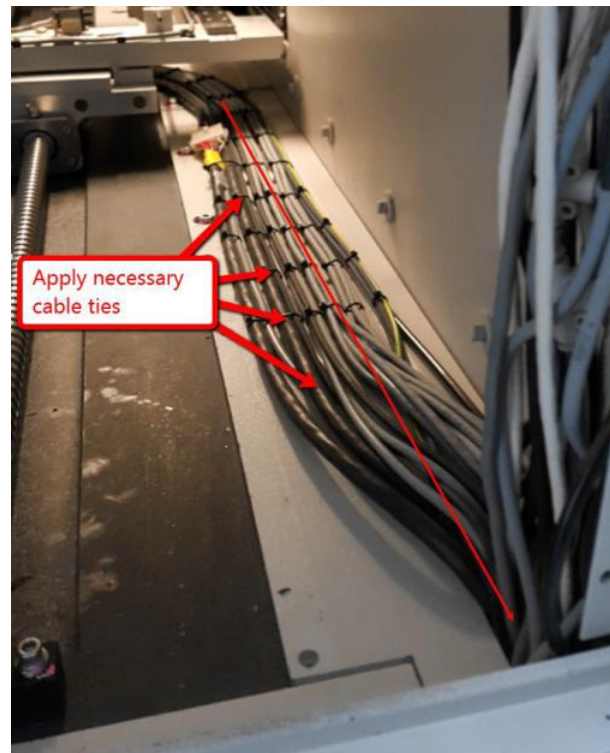


Figure 18 Cable Routing (Read Head Encoder)

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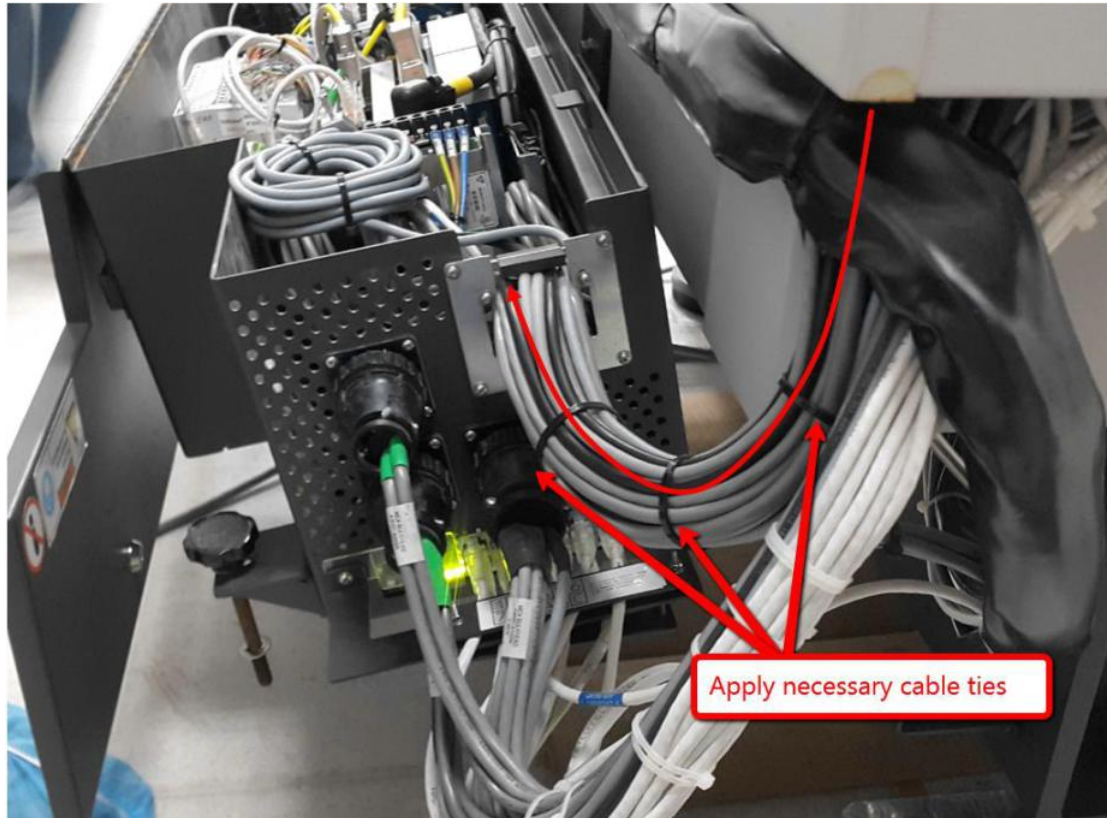


Figure 19 Cable Routing (Read Head Encoder at SSCA)

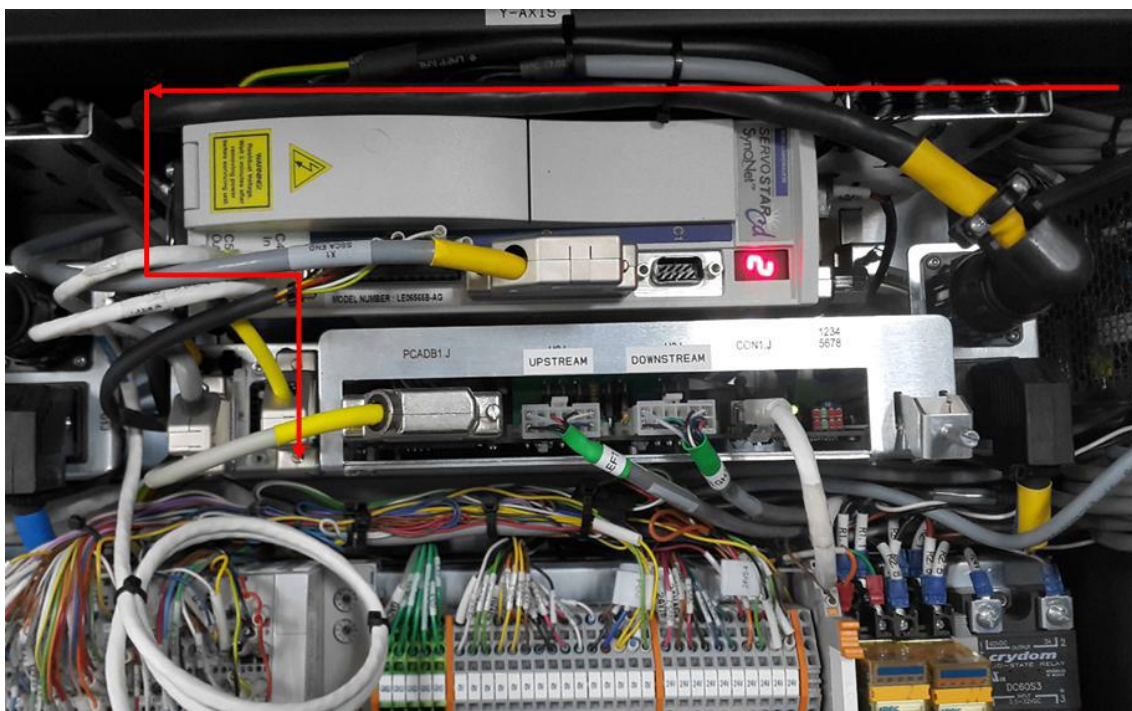


Figure 20 Cable Routing (Read Head Encoder inside SSCA)

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15. Plug in the Y-Axis Read Head Encoder connector from the Y-Axis Kollmorgen as shown in the Figure 21

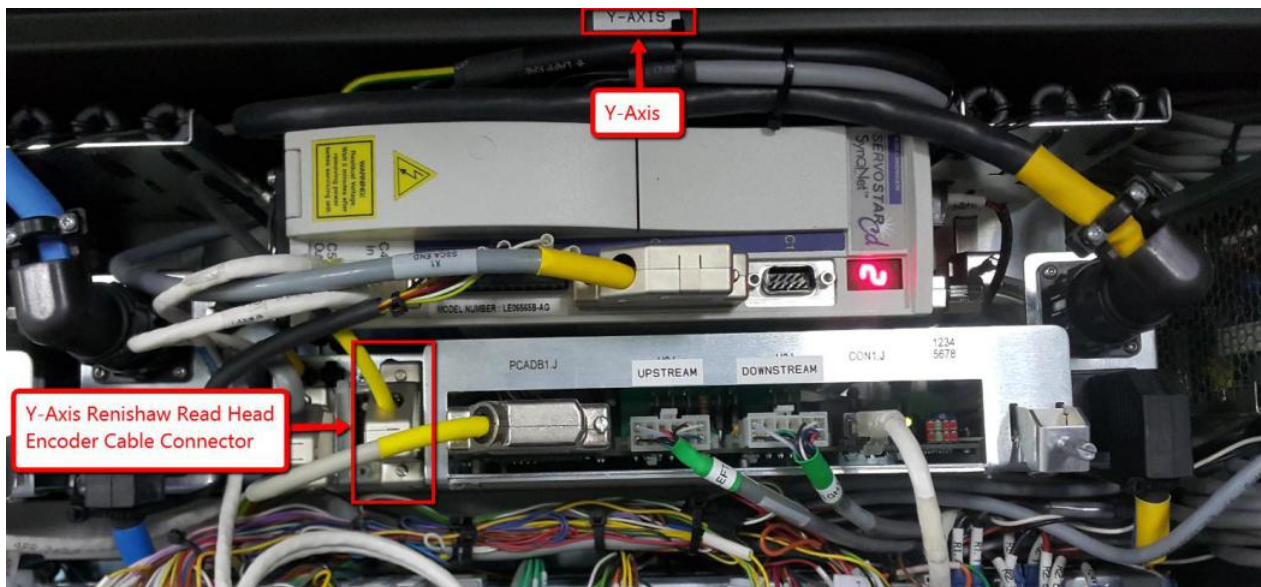


Figure 21 Read Head Encoder Connector

16. Place the cable clamp on its position and tighten back the screw as shown in the Figure 22

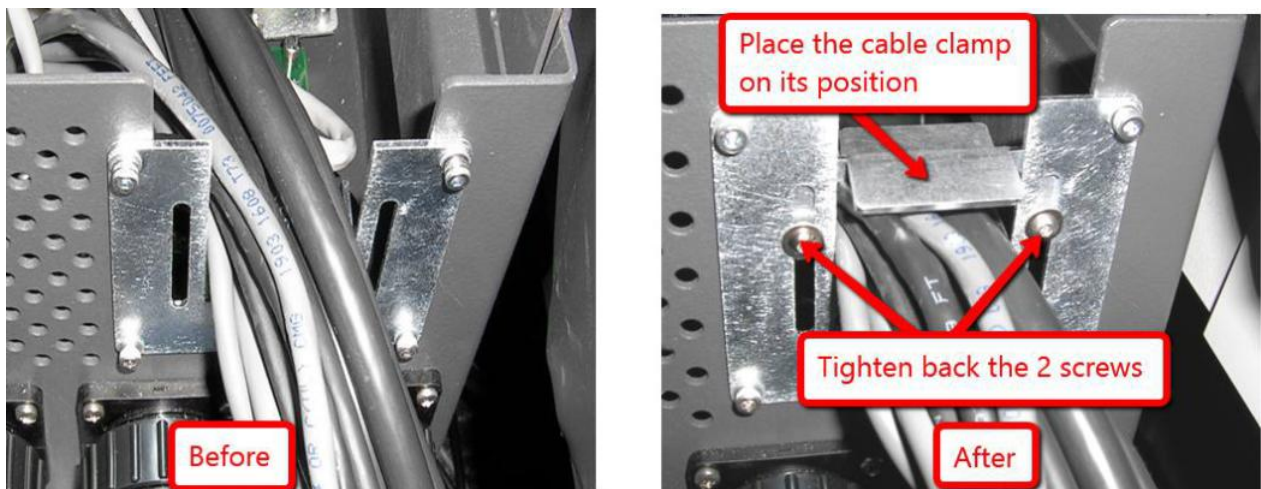


Figure 22 Cable Clamp

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17. Place back the lead cover and push it in the direction shown in the Figure 23. Then, tighten back the 3 screws as shown in the Figure 23

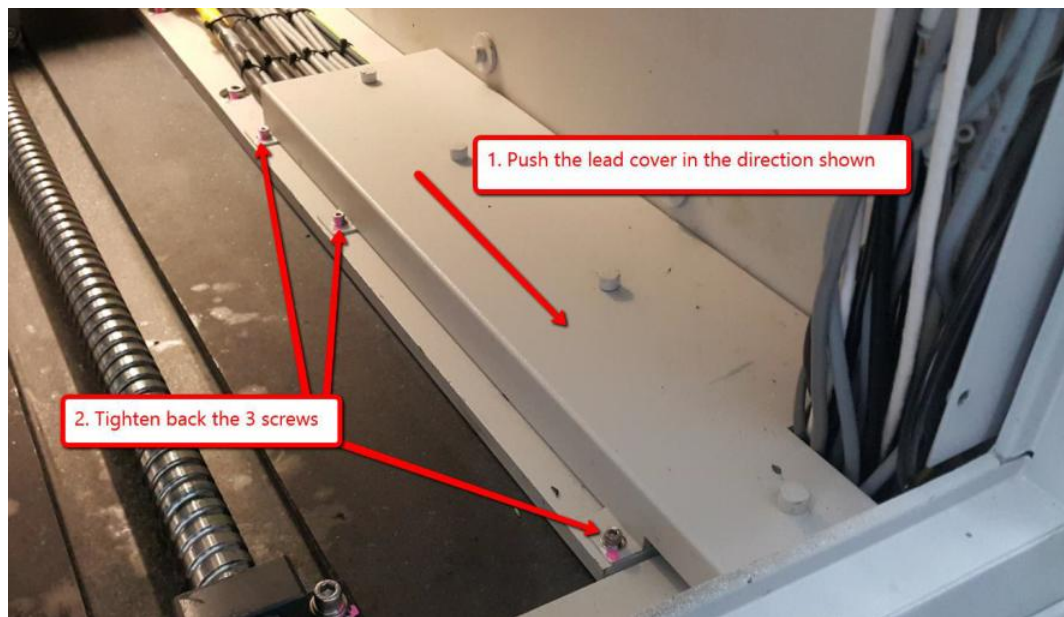


Figure 23 Lead Cover

18. Open the 2 screws and remove the lead cover as shown in the Figure 24.

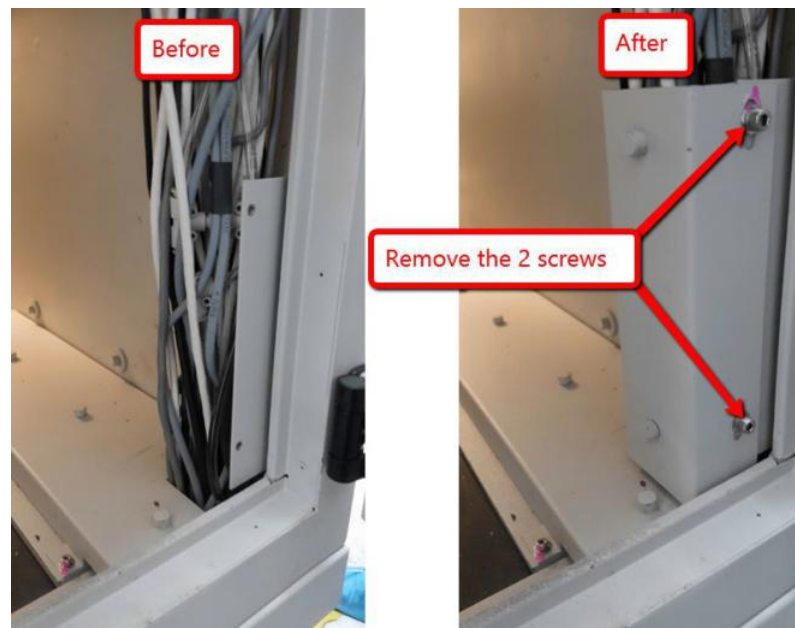


Figure 24 Lead Cover

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19. Once complete, turn on the SSCA and proceed with the alignment of the read head encoder
20. Manually push the stage from the minimum hard stop (Front access door) to the maximum hard stop (Rear access door). The read head encoder should remain Green all the way as shown in the Figure 25. If the read head encoder LED turn red / orange at any location of the stage, continue with the Step 21 for the read head encoder alignment

Remarks: Read head encoder will blink red when detect home sensor



Figure 25 Y-Axis Read Head Encoder (Green)

21. Prepare a Read Head Spacer as shown in Figure 26



Figure 26 Read Head Alignment Spacer

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22. Loosen 2 x screws which shown in the Figure 27

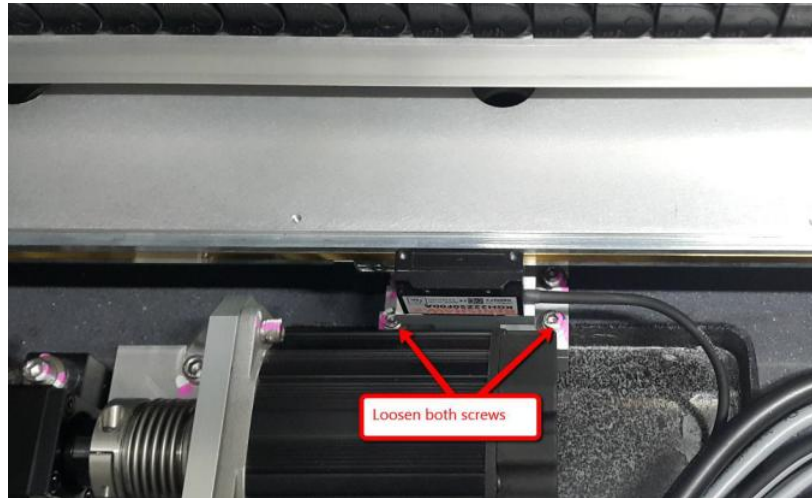


Figure 27 X-axis Read Head

23. Place the read head alignment spacer between the Read Head and the glass scale as shown in the Figure 28.

The cut-out position of the Read Head Alignment Spacer should be positioned over the read head.

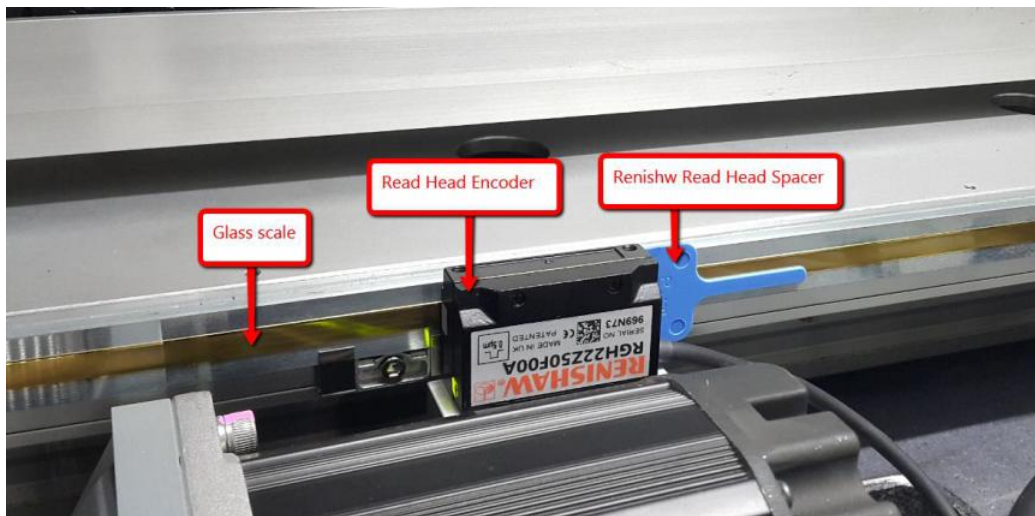


Figure 28 Read Head Alignment Spacer

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24. Push Read Head Encoder towards the Read Head Spacer and tighten the 2 screws as shown in the Figure 29. Once tightened, the spacer should be able to be removed easily from the encoder.

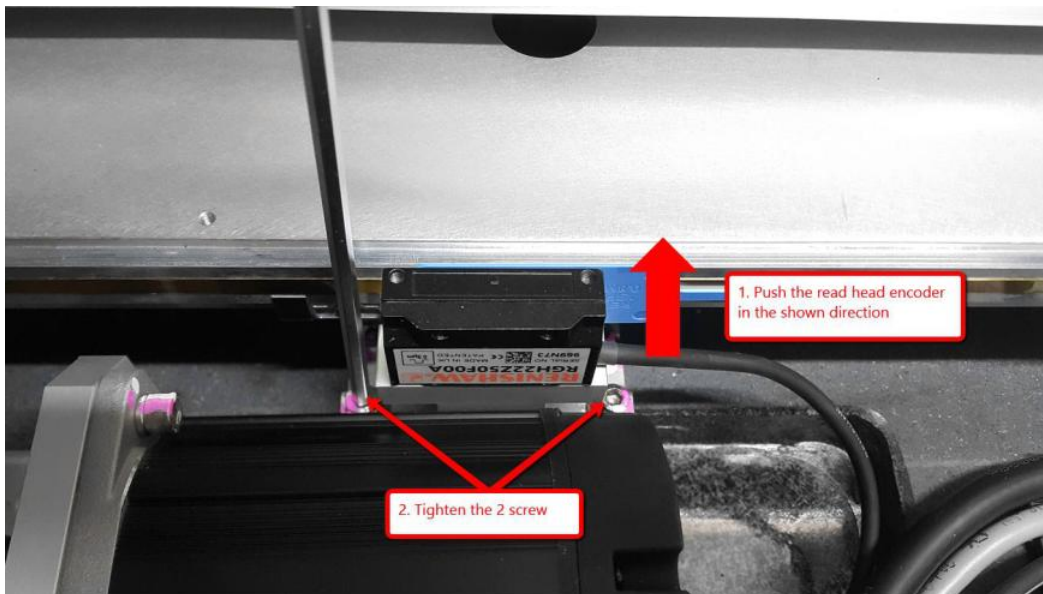


Figure 29 Read Head Encoder

25. Manually push the stage from the minimum hard stop (Front access door) to the maximum hard stop (Rear access door). The read head encoder should remain Green all the way as shown in the Figure 25. If the read head encoder LED turn red / orange at any location of the stage, repeat with the Step 21~25 for the read head encoder alignment

Remarks: Read head encoder will blink red when pass through the limit sensor

26. Assemble/Close the outer barrier and Initialize Panel Positioner tab.
27. Run 5 loops of long scan
28. Run 1 loop of CD & A.